MILITARY SPECIFICATION

GRENADE, HE, G.P., M46
LOADING AND ASSEMBLING

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements and provides quality assurance provisions for loading and assembling of one type of grenade designated as Grenade, HE, G.P., M46.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-13812	-Steel Plate, Wrought, Homogeneous for Ammunition Testing (1/4 to 12 Inches, Incl.)
MIL-E-14970	-Explosive Composition A5
MIL-A-48078	-Ammunition, Standard Quality Assurance Provisions, General Specification For
MIL-G-48226	-Grenades; GP, M42 and M46 Lead Cup Assembly For

FSC 1330

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 -Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-410 -Nondestructive Testing Personnel Qualification and Certification

MIL-STD-1235 -Single and Multi-Level Continuous Sampling Procedures and Tables For Inspection by Attributes

DRAWINGS (See 6.6)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9215370 -Grenade, HE, G.P., M46

INSPECTION EQUIPMENT DRAWINGS

9224981 -Spin Launcher
9225396 -Control Circuits
9202784 -Cap Obturator
9202788 -Operating Instructions

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM E 18 -Standard Test Methods for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to: American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

3. REQUIREMENTS

3.1 Material.- Materials shall be in accordance with the applicable drawings and specifications.

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3.2 Grenades.- The grenades shall comply with all requirements specified on Dwg. 9215370, all associated drawings and with all requirements specified in applicable specifications.

3.3 Static penetration tests of body loading assembly.- The steel block used for the static penetration tests shall be either (1) mild steel, type 1018 or 1020, having a Rockwell hardness of B74 minimum, or (2) steel plate for ammunition testing having material and Brinell hardness in accordance with MIL-S-13812. The penetration requirements of 3.3.1 and 3.3.2 shall apply with the test block oriented such that (a) penetration is perpendicular to the direction of grain of the test block and (b) shaped-charge action impinges upon the rolled surface which is parallel to the direction in which laminations would be formed during manufacture of the test block material.

3.3.1 Normal standoff.- When tested at normal standoff, the body loading assembly, (Dwg. 9215372) shall penetrate a minimum of 3.00 inches of mild steel or a minimum of 2.50 inches of ammunition testing plate.

3.3.2 Modified standoff.- When tested at a standoff distance, which is 4 1/4 + 1/16 inches greater than the normal standoff distance, the body loading assembly shall penetrate a minimum of 5.50 inches of mild steel or a minimum of 4.50 inches of ammunition testing plate.

3.4 Arming test of fuze slide of grenade.- The grenade fuze, less safety clip and safety pin, shall have the arming screw rotated, and held out of the slide permitting the slide assembly to move to the armed position. The armed position shall be considered to have occurred when the distance from the end of the tunnel of the fuze housing to the crown of the beveled end of the slide assembly measures .410 + .035 inch.

3.5 Grenade functioning.- The grenade shall not function prematurely, and shall function high order as intended with evidence of complete shaped charge action.

3.6 Grenade Storage - Grenade bodies removed from the barrier bag packaging and stored as metal parts or partially or fully loaded assemblies, shall not be subjected to a temperature below 50^{0}F (See 6.8).

3.7 First article inspection.- This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

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3.8 Records of results of contractor-conducted examinations and tests.- The contractor shall maintain records of the results of all examinations or tests required by this specification. (See 6.4)

3.9 Workmanship.- All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust and other foreign matter. After loading, all external surfaces, including cavity, shall be free from caked explosive. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous; however, a few light scratches not exposing base material may be permitted. All required marking and stamping shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions.- Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 (PA) shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections.- The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission.- The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part description	Drawings	Quantity
Inert Grenades	9215370	100
Inert Grenades with live fuze	9215370	60
Body Fuze Assemblies	9215371	60
Body Loading Assemblies	9215372	104

4.3.2 Inspections to be performed.- See MIL-A-48078(PA) and Table 1 herein.

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TABLE 1 - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-G-48062C (AR)

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
	Grenade and Subassembly				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Inert Grenade (Dwg 9215370)				
	Unthreading torque test of clinched arming screw	100	-	3.2	4.5.4.2
	Arming test of fuze slide	100	-	3.2	4.5.4.3
	Drop load and torque test of clinched arming screw	100	-	3.2	4.5.4.4
	Pull test of clinched studs	100	-	3.2	4.5.4.5
	Body Loading Assy (Dwg. 9215372)				
	Examination for defects (except 101, 102, and 110-113)	104	-	3.2	4.4.2.4
	Consolidation density and weight of explosive composition	4	-	3.2	4.5.3
	Unseating force of cone swage	50	-	3.2.1	4.5.5.1
	Test for presence of caked explosive entrapped in groove	50	-	3.2	4.5.5.2 or 4.5.5.3
	Body Fuze Assemblies (Dwg. 9215371)				
	Static penetration tests		-		4.5.6
	a. Normal standoff	30	-	3.2	4.5.6.1
	b. Modified standoff	30	-	3.2	4.5.6.2
	Inert Grenade With Live Fuze (Dwg. 9215370)				
	Examination for defects (except 108-112)	60	-	3.2	4.4.2.6
	Functioning	50	-	3.2	4.5.7

NOTES:

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4.3.3 Rejection.- See MIL-A-48078(PA). In addition, if any body fuze assembly fails the normal standoff static penetration test, or if two (2) or more body fuze assemblies fail the modified standoff static penetration test, or if two (2) or more samples fail the functioning test, the first article sample shall be rejected.

4.4 Quality conformance inspection

4.4.1 Inspection lot formation.- Inspection lots shall comply with the lot formation provisions of MIL-A-48078(PA). In addition, inspection lots of Grenades shall contain:

a. Fuzes, M223 of one lot interfix number from one manufacturer.
b. Body assemblies (Dwg. 9215373) of one lot interfix number from one manufacturer.
c. Cones of one lot interfix number from one manufacturer.
d. Tape-stiffener assemblies of one lot interfix number from one manufacturer.
e. Explosive charge material of one lot interfix number from one manufacturer and no more than two serial lot numbers.
f. Lead cup assemblies from not more than two serial lots from one manufacturer.

4.4.2 Examination.- See MIL-A-48078 (PA).

a. Sampling plans.- Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. Continuous sampling in accordance with MIL-STD-1235 may be utilized if approved by the Procuring Activity.

b. Automatic gaging.- Facilities equipped with Government furnished automatic gaging capable of performing 100% inspection of any characteristic shall utilize this capability in accordance with the contract.

c. Alternate inspection procedures.- The contractor may propose the use of alternate inspection procedures in lieu of conducting sampling inspection for defects listed in this specification. (See paragraph 6.10).

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-G-48062C (AR)

PARAGRAPH	TITLE	SHEET OF	DRAWING NUMBER
4.4.2.1	Tape Stiffener Assembly		9215361
			NEXT HIGHER ASSEMBLY 9215370

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Tear, twist or fold in tape		0.40%	3.2	Visual
102	Tape not completely inserted in slots in stiffener		0.40%	3.2	Visual
103	Crack in stiffener		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-G-48062C (AR)

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.2	Pellet (as applicable)				9215248
					NEXT HIGHER ASSEMBLY
					9215372
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Contour of surface interrupted with voids, gouges, chips or broken edges		0.40%	3.2	Visual
Minor 201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET OF	DRAWING NUMBER
4.4.2.3	Body Assembly Type CA Prior to Any Loading (See Note a)		9215373
			NEXT HIGHER ASSEMBLY 9215372

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Internal surface embossed		100%	3.2	Visual or Gage
2	Verificaton of heat treatment		100%	3.2	4.5.1
Major	None defined				
Minor					
201	Yellow marking incomplete or missing		.65%	3.2	Visual or Gage
202	Evidence of poor workmanship		.65%	3.9	Visual

NOTES:

a. These examinations are not required for metal parts originally produced on site and previously inspected for these defects by automatic gage.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET OF	DRAWING NUMBER
4.4.2.4	Body Loading Assembly		9215372
			NEXT HIGHER ASSEMBLY 9215371

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1	Evidence of any crack in body		100%	3.2	Visual or Gage (See 4.5.8)
Major 101	Unseating force of cone swage	Note a		3.2	4.5.5.1
102	Test for presence of caked explosive entrapped in groove	Note a		3.2	4.5.5.2/4.5.5.3
103	Distance from open end to largest cone basic diameter		0.40%	3.2	Gage
104	Distance from largest cone basic diameter to smallest cone basic diameter		0.40%	3.2	Gage
105	Runout of cone at basic length		0.40%	3.2	Gage
106	Largest outside diameter, max.		0.40%	3.2	Gage
107	Lead cup assembly above or excessively below flush		0.40%	3.2	Gage
108	Cone swage missing, incomplete or improper		0.40%	3.2	Visual
109	Any part cracked, dented, distorted or missing		0.40%	3.2	Visual
110	Static penetration test, modified standoff	30/per lot	Note b	3.3.2	4.5.6.2
111	Static penetration test, normal standoff	30/per lot	Note c	3.3.1	4.5.6.1

NOTES: Note a: For each test select 50 samples per lot or 20 samples per machine per eight hours production, accept no failures reject on 1 failure.
Note b: Accept on 2 failures, reject on 3 failures. Note c: Accept on 0 failures, reject on 1 failure.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-G-48062C (AR)

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.4	CONTINUATION SHEET				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
112	Lead cup assembly missing or improperly assembled	Note d	0.40%	3.2	Visual or Gage
113	Aluminum tape missing or improperly applied	Note d	0.40%	3.2	Visual or Gage
Minor 201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES: Note d: inspection performed before loading main charge.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET OF		DRAWING NUMBER
4.4.2.5	Fuze, M223, Prior to Clinching to Body Loading Assembly				9215350
					NEXT HIGHER ASSEMBLY 9215371
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critcal					
1	Depth from bottom of cover to point of arming screw, max.		100%	3.2	Gage
2	Safety pin missing or damaged		100%	3.2	Visual or Gage
3	Slide protruding from fuze tunnel indicating armed or partially armed condition		100%	3.2	Visual or Gage
4	Tabs on cover not fully seated on flange of fuze housing (4 places)		100%	3.2	Visual or Gage
5	Safety clip missing, damaged, or double end not protruding beyond end of fuze tunnel		100%	3.2	Visual or Gage
6	Area of housing tunnel roof containing safety pin hole missing		100%	3.2	Visual or Gage
7	Area of housing tunnel roof which covers detonator missing		100%	3.2	Visual or Gage
Major	None defined				
Minor	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER 9215370
4.4.2.6	Grenade		NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Safety pin of fuze missing or damaged		100%	3.2	Visual or Gage
2	Slide protruding from fuze indicating armed or partially armed conditioned		100%	3.2	Visual
Major					
101	Arming screw not free to rotate		0.40%	3.2	Visual/Manual
102	Evidence of loose or incorrectly positioned tape		0.40%	3.2	Visual
103	Safety clip of fuze missing, damaged or double end not protruding beyond end of fuze tunnel		0.40%	3.2	Visual
104	Arming screw clinch missing, incomplete or not properly centered		0.40%	3.2	Visual
105	Fuze housing cracked or deformed		0.40%	3.2	Visual
106	Stud clinch cracked, split, missing, incomplete or not properly centered		0.40%	3.2	Visual
107	Washer missing		0.40%	3.2	Visual or Gage
108	Unthreading torque test of clinched arming screw	Note a		3.2	4.5.4.2
109	Arming test of fuze slide	Note a		3.4	4.5.4.3
110	Drop load and torque test of arming screw clinch	Note a		3.2	4.5.4.4
111	Pull test of clinched studs	Note a		3.2	4.5.4.5

NOTES: Note a: 2 samples (See 4.5.4.1) per 1800 grenade assemblies produced on each machine, accept on 0 reject on 1 failure.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.6	Grenade				9215370
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
112	Functioning	Note b		3.2	4.5.7/92244981, 9225396, 9202784, 9202788
113	Evidence of gap between fuze housing and cover or fuze cover and grenade cap or loose fuze stake		.40%	3.2	Visual/Manual
Minor 201	Evidence of poor workmanship		.65%	3.9	Visual

NOTES: Note b: 50 samples per lot or 30 samples per 4 hours production, accept on 1 reject on 2 failures.

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4.4.3 Testing.- Testing is required as described herein and in the First Article and Quality Conformance Inspection Table.

WARNING: Some of the following tests are conducted on explosive material and/or assemblies containing explosive components. Test barricades (where used), procedures and equipment shall have appropriate safety approval.

4.4.3.1 Check test for deterioration of lead cup assemblies (See MIL-G-48226) major defect.- If the total time between original acceptance of any lot and the assembly of that lot into the grenades exceeds two years, or if the lead cup assemblies have been subjected to adverse conditions, however brief, at any time since previous tests, the lead cup assembly lot shall be subjected to and must satisfactorily pass, the check test for deterioration specified in MIL-G-48226 immediately before the lead cup assembly lot is assembled into the grenades. This test shall be performed by the contractor on lead cup assemblies selected by the Government Inspector at the facility assembling the lead cup assemblies into the grenades (see 6.7).

4.4.3.2 Moisture content of explosive charge material for body loading assembly (dwg. 9215372) Major defect.- When the explosive material is suspected of having been subjected to adverse storage or shipping conditions permitting the possibility of moisture contamination, a determination of moisture content shall be made. A representative sample of such material shall be selected and tested in accordance with MIL-E-14970. If the moisture content fails to comply with the requirement specified on the applicable drawing, and loading has not begun, the explosive represented by the sample shall not be used for production until corrective action has been taken as verified by repeating this test. In addition, if loading has begun, body loading assemblies with the non-conforming composition shall be rejected.

4.4.3.3 Bulk density of explosive composition (dwg. 9215372) major defect.- The contractor shall provide adequate controls to insure compliance with the applicable requirements. Testing for bulk density will be required only if the Comp A5 is subjected to a subsequent blending, mixing or screening (other than safety) after being received by the loading facility. This testing then shall require that five (5) samples be randomly selected from each such blending, screening or mixing batch and subjected to the bulk density test in accordance with Paragraph 4.5.2. Sampling shall be in accordance with MIL-E-14970. If the bulk density fails to comply with the requirement specified on the drawing, the batch of explosive composition shall be rejected until corrective action has been taken as verified by repeating this test.

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4.4.3.4 Consolidation density and weight of explosive composition of body loading assembly (dwg. 9215372) Major defect. - The density and weight of the explosive charge material of the body loading assembly shall be determined on one assembly from each body loading machine at the start of production each day, and after every one hour of production or portion thereof. If the density or weight of any body loading assembly fails to comply with the drawing requirement, the body loading assemblies produced since the last satisfactory check shall be rejected. In addition, the necessary adjustments shall be made to the machine which will be verified by determining the consolidation density and weight of explosive of two consecutively produced body loading assemblies. Testing shall be in accordance with paragraph 4.5.3.

4.4.4 Inspection equipment.- For the inspection equipment required to perform the examinations and tests prescribed herein the contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078, paragraph 6.3 and paragraph 6.9 herein.

4.5 Test methods and procedures

WARNING: Some of the following tests are conducted on explosive material and/or assemblies containing explosive components. Test barricades (where used), procedures and equipment shall have appropriate safety approval.

4.5.1 Verification of heat treatment of body assembly type CA.- This inspection shall employ the eddy current method, or any other equivalent method approved by the procurement agency. The processes, equipment and calibration procedures used in performing this test, and the hardness limits established shall be documented and have prior written approval of the procurement agency. In addition, after this approval has been granted, the contractor shall make no changes in equipment or procedures without prior written approval of the procurement agency. All equipment used shall be capable of performing this inspection accurately and consistently and shall be calibrated on an approved periodic basis. If the eddy current method is used, all personnel operating and calibrating this equipment shall be qualified and certified in accordance with MIL-STD-410. If any body assembly fails to comply with the requirements specified on the applicable drawing, it shall be classed defective and removed from production.

4.5.2 Bulk density of explosive composition.- The following procedure shall be used to determine the bulk density of the explosive composition:

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BECHTEL METHOD OF BULK DENSITY MEASUREMENT

a. Use 100cc capacity cup (metal cup recommended), polished inside to eight (8) micro-inch finish, and a tare weight equal to the weight of the empty cup. The approximate dimensions of the 100cc cup shall be:

I.D. = 4.0 cm Inside depth = 7.958 cm.
Inside Bottom Radius = 0.076 mm, max.
Wall thickness = 0.2 cm

b. Place both 100cc cup and the tare weight on analytic balance scale, or equal, one on each pan, and check that cup and tare weight balance.

c. Set cup on level surface and fill with composition to be bulk-density tested. To fill, use a scoop, approximately tablespoon size and fill the cup above the rim. Do not tamp or shake cup at any time. Level off excess composition with a straight edge, keeping side of straight edge vertical to rim of cup.

d. Place filled cup on balance pan. Add gram weights to other pan with tare weight until balance is reached. Number of grams added will give the bulk density in percent.

4.5.3 Consolidation density and weight of explosive composition of body loading assembly.- The following procedure shall be used to determine the weight and density of the explosive composition of the body loading assembly. (When weighing in water, use a distilled water containing a suitable wetting agent such as "Aerosol" at 1cc per 250cc distilled water).

a. Prior to assembly, weigh body assembly with lead cup assembly in air (grams).
b. Prior to assembly, weigh the same body assembly with lead cup assembly in water (grams).
c. Dry the components and load with explosive composition using normal production methods.
d. Weigh the assembly in air (grams).
e. Weigh the assembly in water (grams).
f. Calculate the weight of explosive composition in air (d) - (a).
g. Calculate the weight of explosive composition in water (e) - (b).
h. Calculate weight (and volume) of water displaced by the explosive composition (f) - (g).
i. Calculate the density of loaded explosive composition (f)÷(h) grams/cc.

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The body assemlies with lead cup inserted shall be randomly selected prior to loading and identified as test samples. Each sample shall be marked clearly with its weight and the applicable powder consolidation machine number. The samples must be retrieved after powder consolidation but prior to cone insertion.

NOTE: Contractors utilizing the cone for re-consolidation of the explosive shall develop a procedure for performing this test. The proposed procedure shall be submitted to the address specified in par. 6.4 for approval, prior to use.

4.5.4 Fuze to grenade assembly test

4.5.4.1 Preparation of sample items.- The samples to be subjected to this inspection shall consist of a fuze, a tape stiffener assembly, and an empty body assembly from lots used in production of the grenade inspection lot. (See 4.4.1) They shall be prepared at the specified times from each assembly machine used during lot production. The samples shall be suitably marked to prevent mixing with grenades not intended to be samples, and to allow identification of the particular assembly machine on which the unit was produced.

4.5.4.2 Unthreading torque test.- The sample shall be mounted in the approved fixture. The safety clip and the safety pin shall be removed. With the fixture preventing the fuze slide from arming, the torque required to rotate the arming screw out of the slide shall be measured. If the required torque exceeds the amount specified on the applicable drawing, the sample shall be classed defective.

4.5.4.3 Fuze slide arming test.- After completion of the unthreading torque test, while maintaining the arming screw free of the slide, the fixture device for preventing slide movement shall then be withdrawn permitting the slide to move to the armed position. After observation for compliance with the arming requirements and while still maintaining the arming screw free of the slide, the slide shall be returned to the unarmed position. The safety pin and safety clip shall be properly reinserted in the fuze. Grenade fuzes failing to comply with the specified arming distance, shall be classed defective.

4.5.4.4 Drop load and torque test of arming screw clinch of grenade fuze.- After completion of the arming test, the same sample shall be mounted in an approved fixture and subjected to the drop load as specified on the applicable drawing. The crimp of the tape stiffener assembly to the arming screw shall the be subjected to the torque as specified on the applicable drawings. If the tape rotates relative to the arming screw, when the specified torque is applied, the assembly shall be classed defective.

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4.5.4.5 Pull test of clinched studs of body fuze assembly.- After completion of the drop load and torque test of the arming screw clinch the same sample assemblies (with safety clip and safety pin properly inserted in the fuze) shall have the tape stiffener assembly removed. This may be accomplished by any method which will not damage the arming screw threads. After removal of the tape stiffener assembly, the safety clip only shall be removed and the arming screw rotated to the armed position. The safety clip shall then be reinserted sufficiently to prevent the arming screw from freely rotating in the fuze while a suitable adapter is assembled to the threads on the protruding end of the arming screw. The sample item shall then be assembled to an approved fixture and the load specified on the applicable drawing shall be applied to the clinched studs by pulling on the assembled adapter. If the fuze loosens as evidenced by a gap between the body assembly and the fuze, the sample assembly shall be classed defected.

4.5.4.6 Disposition.- If any body assembly fails to comply with any of these tests the production quantity represented by that sample shall be rejected.

4.5.5 Unseating force test of cone swage and test for presence of entrapped powder of body loading assembly - The force test and test for entrapped powder may be combined provided the contractor uses powder nests or equivalent loading aids to reduce the occurrence of entrapped powder during grenade body loading. To use this alternative the contractor must, prior to use, request approval from the procuring agency and furnish documentation of the method planned to reduce the possibility of entrapped powder. For this combined test the same samples are used for both the force test and entrapped powder test. After having passed the 2000 lb. force requirement, the load shall be increased until the cone is dislodged from the grenade body, the grenade body and the surrounding area on the test fixture shall be examined for evidence of powder that had been entrapped between the cone and the grenade body. Any evidence of entrapped powder shall be cause for rejection of that body loading assembly.

4.5.5.1 Preparation of sample items and test procedure for unseating force - The sample assemblies shall be randomly selected from normal production.

The quantity of body loading assemblies shall be cut in two in a plane perpendicular to the longutidinal axis of the body. The cut shall be located at a minimum distance of .800 inches from the open end. The cut shall be made cleanly through the body wall and explosive material.

The portion of the cut sample containing the swage of the cone shall have any remaining explosive material cleaned away and then shall be mounted in an approved fixture. The load as specified on the applicable drawings shall be applied to the cone. If the cone is dislodged from its swaged position in the body loading assembly by the specified load, that body loading assembly shall be classed defective.

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4.5.5.2 Test for presence of entrapped powder in body loading assembly.- The sample assemblies shall be randomly selected from normal production. The quantity of body loading assemblies shall be cut on the longitudinal axis in two halves. The cut shall be made cleanly through the body wall, the explosive material, and cone. The sectioned cone in each half shall be lifted out and visual examination shall be performed to determine compliance with the requirement specified on the applicable drawing. Any evidence of entrapped powder shall be cause for rejection of that body loading assembly.

4.5.5.3 Test for presence of entrapped powder in body loading assembly (alternative method).- The sample assemblies shall be randomly selected from normal production. The quantity of body loading assemblies shall be cut in a plane perpendicular to the axis of the body at a minimum distance of .800 inches from the open end. The cut shall be made cleanly through the body wall, cone and filler material.

The ring containing the base end of the cone shall then be cut into two semi-circular halves so that the cone can be lifted out. An alternative to the above method is to cut the cone into two semi-circular halves leaving the base end of the body in ring form. The cone halves shall then be removed but without loss of any entrapped powder.

The two halves of the cone shall be lifted out and the body and cone halves shall be visually examined for any evidence of caked powder entrapped in the groove or between the body wall and the skirt of the cone. Any evidence of entrapped powder shall be cause for rejection of that body loading assembly.

4.5.5.4 Disposition.- If any body loading assembly fails to comply with any of these tests the production quantity represented by that sample shall be rejected.

4.5.6 Static penetration tests of body loading assembly.- The initiating device for these tests shall be the M223 fuze, (Ref. Dwg. 9215350). The loaded body and fuze may be assembled off the production line provided the procedures and general techniques utilized reproduce the techniques of the production line.

The test item shall be taken to a suitable test area for performing these tests. Prior to arming the fuze, the test block shall be oriented such that penetration will occur perpendicular to the direction of grain of the test block and shaped charge action will impinge upon the rolled surface which is parallel to the direction of laminations in the block. The test item shall be placed on the test block. The fuze shall be armed by removing the safety clip, (Ref. Dwg. 9215356) but not the safety pin, (Ref. Dwg. 9215255) and unthreading the arming screw out of the weight (Ref. Dwg. 9215355). The arming screw shall be maintained free of the

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slide by inserting a suitable blocking device between the fuze slide and the arming screw point. The safety pin shall then be removed permitting the slide to move to the armed position. The blocking device shall be removed from a remote location behind an approved barricade. The blocking device shall remain in place until immediately prior to dropping the weight.

The body loading assembly shall be initiated by using fixturing which will permit a 2-ounce weight to drop from a height of 4 inches above the top of the arming screw onto the armed fuze. The falling weight shall then drive the firing pin (on the end of the arming screw), into the detonator to initiate the firing train.

If the initiating device fails to function, the test shall be considered invalid, the test unit shall be destroyed in place using approved procedures, and a replacement item shall be tested in its place to complete the sample for these tests.

The tests using normal standoff shall be conducted as specified in 4.5.6.1 and the tests using modified standoff shall be conducted as specified in 4.5.6.2. All tests shall be conducted remotely from behind approved barricades. All fixturing and operating procedures utilized shall be approved.

NOTE: Short test blocks suitable for individual shots may be used in lieu of the 18 inch long test blocks specified in 4.5.6.1 and 4.5.6.2 for multiple shots. However, the size of the test blocks for individual shots shall not be less than 2 1/2 inches X 2 1/2 inches, shall be of material and height in accordance with 3.3 and shall be oriented such that penetration is in the direction specified above.

4.5.6.1 Normal standoff.- This test is required as part of the First Article Testing. Subsequent testing for this requirement shall be only when directed by the Contracting Officer. The block for these tests shall be of material and height in accordance with 3.3, and shall measure 2 1/2 + 1/16 inches wide and a minimum of 18 inches long. The block shall be placed on a firm horizontal surface with the 2 1/2 inch width uppermost and the body loading assembly placed on it so the center line on the cone is 2 inches from the end (end A). Each subsequent shot is to be located a distance of 2 inches from the center line of the previous hole and the center line on the last (eighth) shot for that surface shall be located a minimum of 2 inches from the opposite end. The block shall then be turned over (180°) and the center line of the cone of the next body loading assembly shall be located 3 inches from the end (end A). Each subsequent shot is to be located a distance of 2 inches from the center line of the previous hole and the center line of last (fifteenth) shot shall be located a minimum of 3 inches from the opposite end.

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4.5.6.2 Modified standoff.- Each body loading assembly to be subjected to this test shall be set into a tube or other approved holding device such that the normal standoff distance is increased by the modified standoff distance (See 3.3). The block, of material and height in accordance with 3.3, shall measure 2 1/2 + 1/16 inches wide and a minimum of 18 inches long. The block shall be placed on a firm horizontal surface with 2 1/2 inch width uppermost and the body loading assembly (with increased standoff) placed on it so the center line of the cone is 2 inches from the end (End A). Each subsequent shot is to be located a distance of 2 inches from the center line of the previous hole and the center line of the last (eighth) shot for that surface shall be located a minimum of 2 inches from the opposite end. The block shall then be turned over (180°) and the center line of the cone be located 3 inches from the end (end A). Each subsequent shot is to be located a distance of 2 inches from the center line of the previous hole, and the center line of the last (fifteenth) shot shall be located a minimum of 3 inches from the opposite end. The depth of shots failing to penetrate the entire block height shall be measured with a depth gage or other probe. Should any measurement fail to comply with the specified minimum depth requirement, and lot acceptance is dependent upon the result of that shot, the depth shall again be measured, using radiographic means.

NOTE: The above is a recommended test procedure. Requests for the use of alternate procedures shall be submitted to: Commander, US Army ARRADCOM, ATTN: DRDAR-QAR-Q, Dover, New Jersey 07801, for approval prior to use.

4.5.7 Grenade functioning.- The inert grenade, with live fuze and with safety pin (Dwg. 9215255) and safety clip (Dwg. 9215356) removed, shall be fitted with two obturators (Dwg. 9202784) and loaded into an approved launcher. While spinning at a minimum rate of 5000 rpm, (max rate not to exceed 5500 rpm), the grenade shall be launched to impact upon a target located at a minimum horizontal distance of 100 feet (not to exceed 150 feet) from the launcher. The velocity (average) of each grenade at target impact shall be 130 ± 20 feet/second. The target shall be steel and have a thickness of 3/4 + 1/16 inch. This target shall be firmly supported in a plane perpendicular to the line of fire. For each grenade lot tested, the contractor shall furnish proof of meeting the requirement specified herein to the satisifaction of the contracting officer. Any definite evidence of equipment malfunction which causes a failure to occur, shall be just cause for replacement of that shot in the sample.

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After each grenade is fired, observation shall be made for evidence of fuze arming and detonator functioning. In addition if it is determined that during the test the detonator did not function, due to improper attitude of the grenade at target impact, the results of that grenade shall be invalidated and a replacement fired. The inert body loading assemblies used for this test shall be suitably marked and randomly introduced into the body fuze assembly machines in production so as to be representative of the grenades for which the acceptance is being determined by this test. If any grenade fuze fails to arm or function high order it shall be classified defective.

4.5.8 Inspection of body for cracks.- This inspection shall be performed on each body after loading either visually or with an approved NDT method such as Eddy Current. If the eddy current method is used, all personnel operating and calibrating this equipment shall be qualified and certified in accordance with MIL-STD-410. Inspection shall be performed at each of the following locations.

a. Longitudinal defects in the wall of the body from the open end to the shoulder area.

b. Circumferential defects in the cap area of the body originating from the exterior surface.

4.5.8.1 For the purpose of identifying a reject, the following definition of a rejectable crack will apply:

a. Crack - A discontinuity in the material with an orientation, depth, and length represented by approved standards.

5. PACKAGING.- (This section not applicable to this specification).

6. NOTES

6.1 Intended use.- The component covered by this specification is intended for use with the Projectile, 155MM, HE, M483A1.

6.2 Ordering data.- See MIL-A-48078 (PA).

6.3 Submission of inspection equipment designs for approval.- See MIL-A-48078 (PA). Submit equipment designs as required to: Commander, US Army ARRADCOM, ATTN: DRDAR-QAR-I, Dover, New Jersey 07801.

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6.4 Data requirements.- Data item descriptions required for submission under this specification are as follows:

Paragraph	Data requirement	Applicable D.I.
4.5.1, 4.5.4, 4,5,5, 4.5.7, 4.5.8	Quality Inspection Defect Report	DI-R-1721
4.5.3, 4.5.6	Quality Inspection Test Demonstration and Evaluation Report	DI-R-1724
6.10	Procedure Report (Tailored)	*DI-T-5372

*Delete a, b, c, d of paragraph 1 block 10, DD Form 1664, and substitute contractor designated parameters and limits.

The above data shall be forwarded in accordance with the requirements of the contract to Commander, ARRADCOM, ATTN: DRDAR-QAR-Q, Dover, NJ 07801.

6.5 Barricades.- Proper barricades shall be utilized for all inspections performed, when applicable.

6.6 Drawings.- Drawings listed in Section 2 of this specification under the headings US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.7 Cost of check test.- The contracting officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of the check test for deterioration of the lead cup assemblies. The tests shall be conducted at Government expense without cost to the contractor who loaded the lead cup assemblies, or to the contractor assembling the lead cup assemblies into the grenades; and shall not constitute a basis for rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the lead cup assembly lot was under the jurisdiction of either contractor.

6.8 Grenade storage.- The elapsed time between the loading and assembling of the grenades and their assembly into the projectile shall be kept to a minimum.

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6.9 Automatic inspection equipment.- All automatic test and inspection equipment when utilized by the contractor shall be subjected to a periodic verification and/or a calibration check at frequent intervals during normal production. The procedures shall include as a minimum: frequency, method of verification, and the method of retrieval of items produced between verification/ calibration. These procedures shall be included in the operation/calibraiton procedures as required by paragraph 4.4.4.

6.10 Alternate inspection procedures.-

6.10.1 Justification for alternate inspection procedures.- Submitted documentation must include sufficient information to indicate an economic benefit, and must also clearly identify that there is no reduction in product quality.

6.10.2 Alternate inspection procedures.- The alternate inspection procedures as a minimum must include:

a. Frequency (producton interval) for validating equipment functioning.

b. Proposed method of validation to include process and inspection drawings.

c. Description of material control procedures to be imposed during the production interval between validations (Note: Under these provisions final acceptance will be withheld from material produced during a production interval until conducting satisfactory validation.)

6.10.3 Approval of alternate inspection procedures.- Proposed plans shall be submitted per para. 6.4 for approval.

6.11 Submission of data cards.- In addition to the normal distribution of data cards, when the grenade is procured by the US Army Armament Materiel and Readiness Command, one (1) copy of all ammunition data cards shall be forwarded to: Commander, USA ARRADCOM, ATTN: DRDAR-QAR-Q, Dover, NJ 07801.

Custodian:
Army-AR

Preparing Activity:
Army-AR

(Project 1330-A169)

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